

**Networking and Information Technology Research
and Development (NITRD)
Office of Science and Technology Policy (OSTP)
Development of an Artificial Intelligence (AI) Action Plan**

Request for Information (RFI)

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1 Executive Summary

The United States stands at a pivotal moment in the global Artificial Intelligence (AI) landscape. While it has long been a leader in technological innovation, rapid advancements by peer competitors, shifting industry dynamics, and regulatory uncertainties require a proactive and comprehensive national strategy. The United States has historically led global technological innovation through strategic government policies that prioritize education, research and development, public-private partnerships, and market-driven competition.

To sustain and enhance this leadership in AI, the forthcoming AI Action Plan must incorporate concrete policy that align with successful historical precedents while addressing contemporary challenges. Conversely, a lack of U.S. leadership in developing clear AI standards would create an international vacuum, allowing the EU, China, or other economic rivals to develop de facto global standards, providing foreign companies an economic edge. In turn, we offer key policy recommendations across major AI policy areas. By implementing the proposed policy actions focused on education and workforce development, research and development, hardware, AI risk management, national security applications, and fostering innovation through public-private collaboration, the U.S. can reinforce its leadership in AI and establish long-term competitiveness.

Ardent Management Consulting, LLC (Ardent) has provided digital transformation, location intelligence, and analytics to numerous government agencies for over 18 years. Ardent is all in when it comes to providing exceptional service and quality solutions to our customers. We are a proud service-disabled veteran owned business striving to improve the mission and support our nation. The technology and threat landscape are ever evolving, and we have taken significant strides to stay abreast by standing up our own innovation unit and pouring resources into developing our intellectual and human capital with regards to artificial intelligence and machine learning. As a company competing in the Artificial Intelligence/Machine Learning (AI/ML) market and driving policy through National Institute of Standards and Technology's (NIST's) AI Safety Institute Consortium, we have firsthand knowledge of the opportunities and challenges faced in the industry and we are committed to supporting our nation to the best of our ability.

History has shown that strategic government intervention, whether through investment in foundational research, regulatory adaptability, or fostering public-private partnerships, has played a key role in advancing technology. The AI Action Plan must continue this tradition by aligning federal policies with the needs of academia, industry, and national security. A well-structured, forward-looking approach will not only drive AI innovation but also create a resilient ecosystem that ensures economic growth, national security, and responsible AI deployment.

By making AI a national priority—one that is accessible, reliable, and innovative—the U.S. can shape the future of AI development and application, setting global standards while maintaining its leadership in the most transformative technology of our time.

2 Relevant AI Policy Topics

2.1 Hardware and Chips

Suggested Policy Action: The government must put forth policy to spur continued improvement in AI hardware, including GPUs, ASICs, and novel computing architectures like neuromorphic, reversible, and quantum computing.

Policy Areas: Grants for fundamental and applied research, SBIR and transition funding for commercialization, CRADAs for public-private partnerships, and tax incentives for performing innovative work.

Intended Outcome: Establish a full-lifecycle support system that encourages continuous technological growth, maintaining the U.S. lead in AI hardware advancements and preventing adversaries from surpassing domestic capabilities.

The third AI summer started in 2012, when the first deep neural network (AlexNet) was successfully trained on a Graphics Processing Unit (GPU) and significantly outperformed the competition on the ImageNet challenge. Since then, GPUs, driven by PC gamers and AI/ML researchers, have grown exponentially in performance. The recent Framework for Artificial Intelligence Diffusion was implemented to keep these powerful GPUs out of the hands of our nation's adversaries and so far, this effort has been moderately successful. This has been demonstrated by the release of DeepSeek r1, which took the world by storm when it was announced that it trained on a fraction of the compute that other leading foundation models require.

However, since GPUs are a commodity, completely preventing the flow of them to other countries, both directly and indirectly, is virtually impossible. In the private sector, blocking others is only an effective strategy for monopolies, everyone else must compete and take efforts to maintain their competitive advantage. The same can be said of nations. Rather than ineffectually blocking other countries' technological progress, we must fight to maintain our technological edge.

Therefore, the government must put forth policy to spur continued improvement in hardware for AI, such as GPUs, Application Specific Integrated Circuits (ASICs), and novel computing architectures like neuromorphic, reversible, and quantum computing. The policy must include grants for fundamental and applied research, Small Business Innovation Research (SBIR) and transition funding for commercialization, establish Collaborative Research and Development Agreements (CRADAs) for public-private partnerships, and tax incentives for performing innovative work, to provide a full-lifecycle support system that encourages continuous growth in this area.

2.2 Education and Workforce Development

Suggested Policy Action: Establish a federal AI-specific workforce training program similar to the National Defense Education Act of 1958, develop robust continuing education programs for government and private sector employees, and create public-private partnerships to offer free educational resources on AI.

Policy Areas: AI education initiatives, federal workforce upskilling, industry collaboration, and community outreach programs.

Intended Outcome: Ensure the U.S. maintains its leadership in AI talent, bridge the hiring gap, enhance workforce productivity, and prevent the U.S. from falling behind competing nations.

Intellectual capital is the foundation of any knowledge-based industry. Unfortunately, the United States' lead in AI talent has eroded. A study done by MacroPolo (associated with the Paulson Institute, a non-profit organization focused on U.S.-China relations) on global AI talent sampled papers from the Conference on Neural Information Processing Systems (NeurIPS) 2022 and collected information on the authors' country of origin, where they studied, and where they currently work. They found that between 2019 and 2022 the most elite global AI talent from the U.S. had slipped from 35% of the total to 28%, barely ahead of China's 26%. At the rate of growth China exhibited in 3 years, another study would likely show their share to have surpassed that of the U.S.

To build and maintain our nation's lead in global AI talent, we must create a federal AI-specific workforce training program, similar to the National Defense Education Act of 1958. In addition, a robust continuing education program for all other government and private industry employees would enable workers to better understand and use AI to augment their productivity and job outcomes. Studies show that up to 70% of workers lack the AI skills needed to fully use AI tools to the best of their capability. With the growing adoption of AI, without this workforce training program, employers would need to spend resources to upskill both new hires and legacy employees – resources they may not opt to spend, which would overall decrease our workforce's dominance in the global landscape.

This staggering figure reveals an unparalleled opportunity to unleash a new wave of productivity and prosperity if we fully embrace AI throughout our entire workforce. The alternative is if we do not upskill our workforce and other nations do, we will be left behind. Finally, for those not in the workforce, as part of this training program, we suggest forming a public-private partnership with academia and leading AI companies to develop free resources and seminars to educate the community on practical applications and address common misconceptions about AI.

2.3 Research and Development (R&D)

Suggested Policy Action: Establish national AI research centers under the NSF, expand funding for fundamental and applied AI R&D, and incentivize continued private sector investment through expanded R&D tax credits, particularly for small businesses.

Policy Areas: AI research funding, public-private partnerships, innovation tax incentives, and risk-tolerant RDT&E strategies.

Intended Outcome: Stabilize long-term AI R&D funding, buffer against short-term market fluctuations, foster cutting-edge AI innovations, and ensure continued U.S. dominance in AI advancements.

The AI arms race is in full force, and R&D is the engine driving it. The Human-centered Artificial Intelligence Institute at Stanford University tracks the industry with their AI Index. Their 2024 report showed that the U.S. led in multiple measures, such as the number of notable models released, however, this is no reason to rest on our laurels; highly innovative industries such as the AI industry are extremely volatile. Any slowdown in R&D could have significant consequences in terms of global technical leadership.

Private industry's investments in AI have exceeded government funding, presenting a risk to the nation's R&D output stability. Publicly traded for-profit businesses operate on short-term cycles

and are beholden to shareholders, meaning funding could rapidly diminish if rates of return are poor for a few quarters. For instance, experts have argued that the current technology used in large language models (LLMs) is fundamentally incapable of higher-level reasoning and planning since the models hallucinate and have no understanding of the world; if consumers become disillusioned with these LLMs and agents made with them, revenue, and in turn R&D funding, would dry up.

The government on the other hand has different goals and incentives and could act as a buffer to large oscillations in R&D funding and efforts. There are a few things the government could do to act as this buffer and continue to spur growth from R&D:

- The U.S. should first establish national AI research centers under the National Science Foundation (NSF) to drive collaboration between academia, industry, and government.
- The U.S. should greatly expand funding for fundamental and applied AI R&D, as well as enabling technologies.
- The U.S. should incentivize continued private sector investment in AI innovation through expanded R&D tax credits, especially for small businesses.

Innovation is inherently risky; it is the act of doing something that has not been done before. Silicon Valley as a collective has a high-risk tolerance, as companies are formed and dissolved daily in pursuit of the next billion-dollar business. The government has a loftier pursuit to steward taxpayer funds. We do not suggest that the government take the stance of Silicon Valley and bet everything on a single endeavor, however, without risk, there cannot be reward.

Instead, we suggest the organizations managing Research, Development, Testing & Evaluation (RDT&E) funds endeavor to develop a risk tolerant culture that acknowledges failure is an option to drive our collective understanding, enable revolutionary innovation, and improve our nation. By doing so, the U.S. will realize a greater return on investment from pursuing such innovations, offering higher rewards compared to less risky incremental innovations.

2.4 Risks, Regulations, Explainability and Assurance of AI Model Outputs

Suggested Policy Action: Use NIST's AI Risk Management Framework as a base to develop automated tools and systems for AI model test and evaluation, explainability, and regulatory compliance. Establish a consortium with IEEE or ISO to develop AI industry-wide standards.

Policy Areas: AI model assurance, open-source compliance tools, risk-based regulatory framework, and industry standardization.

Intended Outcome: Reduce compliance burdens while ensuring AI reliability and transparency, prevent the emergence of "shadow AI," and foster responsible AI adoption within government and industry.

Reducing regulatory burden has been shown to enable innovation, such as when the Federal Communications Commission (FCC) designated the 2.4GHz band for unlicensed use and Wi-Fi and Bluetooth expanded. However, there is a tradeoff between regulation and reliability, especially when there is a risk of harm to humans.

We suggest a solution where, using NIST's AI Risk Management Framework as a base, the government invests in automated tools and systems for test and evaluation of model outputs and explainability to reduce the burden of compliance, while enabling reliable AI systems. These tools should be open-source and available to model developers and integrators to run on their

own as needed to comply with any industry-specific regulations. Ardent has been developing such a collection of tools for the past 18 months.

Furthermore, we suggest creating a consortium with the Institute of Electrical and Electronics Engineers (IEEE) or the International Organization for Standardization (ISO) to develop industry-wide standards for products and services that leverage AI to improve consumer confidence. Rather than banning AI's use among end users, we recommend the government lead AI technology adoption across its ranks, enabled by contractors who are leaders in AI technologies like Ardent.

For example, consider an AI used in emergency response that incorrectly predicts a catastrophic event, leading to unnecessary evacuations or delayed critical aid. To address this, technical standards for prediction accuracy are not enough – we need clear accountability structures to define who is responsible when an AI miscalculation leads to public panic or resource misallocation. We suggest establishing governance that clarifies liability, ensures transparency in algorithm training, and builds public trust through thorough and independent audits. This requires collaboration with emergency management professionals, AI developers, and ethicists to develop comprehensive policies that align with both national and international best practices, ensuring the responsible enhancement of public wellbeing.

We urge the government to recognize its users are adopting AI whether permitted by their respective agencies or not. By being at the forefront of adoption, the government can observe to learn the risks, then work to mitigate them. Without being aware of these risks, agencies will suffer from invisible adoption or “shadow AI” where the risks are unknown because they cannot be observed.

2.5 National Security and Defense Applications

Suggested Policy Action: Adopt an AI-first strategy for government IT infrastructure, modernize acquisition processes to enable faster AI adoption, expand SBIR funding, and increase follow-on funding opportunities for small businesses developing AI capabilities.

Policy Areas: AI-driven IT modernization, defense acquisitions reform, federal AI adoption incentives, and national security R&D investments.

Intended Outcome: Enhance national security through AI-driven operational efficiency, foster defense innovation, and stimulate economic growth by leveraging small business contributions to AI advancements.

The government's information technology (IT) strategy and infrastructure has fallen well behind private industry because critical legacy systems are built on outdated frameworks. However, this presents an opportunity: the government can leapfrog private industry and our peer adversaries by taking an AI-first approach to our nation's IT strategy. While there are clear risks, contractors take the brunt of the risk by investing in their own R&D to then present finished, well-researched AI products and technologies to the government. By putting AI front and center it will require agencies to adapt policies and technologies to rapidly adopt and field AI systems.

Part of that policy change will require lowering the barrier to entry of non-traditional contractors, enabling faster adoption and integration of new and innovative technologies. For example, many Microsoft licenses already include access to generative AI through Microsoft CoPilot, meaning that by not adopting its use the government loses money on each employee's individual Microsoft license. By extension, putting AI in the hands of every government employee will spur

productivity, further innovation as users discover new ways to use AI, and create use-cases that will require further R&D. This process invigorates the technology innovation cycle and further reinforces the U.S. as an AI technology powerhouse.

Furthermore, the government should expand the SBIR program. In addition, the government should also expand follow-on funding opportunities such as the Strategic Funding Increase (STRATFI) and Tactical Funding Increase (TACFI) Programs, allowing small businesses to flourish while meeting government needs. For example, as previously mentioned, wide adoption of AI will bolster innovation and create new opportunities as users develop new use-cases requiring further R&D. The DoD SBIR National Economic Impacts, 1995 – 2018 report shares that over this period, the DoD, which funded approximately 50% of all SBIRs during that time, saw a return on investment of \$22 for every \$1 spent, with a total economic impact of \$347B. Expanding the SBIR program would clearly create a win-win scenario where the nation's AI capabilities are advanced while simultaneously improving the economy.

2.6 Innovation, Collaboration, and Competition

Suggested Policy Action: Establish a public-private AI supercomputing center to provide small businesses and research institutions with access to high-performance computer resources for AI training and experimentation.

Policy Areas: AI infrastructure investment, government-industry-academia collaboration, and democratized access to AI resources.

Intended Outcome: Level the playing field for AI innovation, enable groundbreaking research beyond large corporate entities, and strengthen U.S. global competitiveness in AI development.

A significant limitation of small businesses and research institutions is that they do not have access to the sheer amount of computing power or capital that leading AI firms possess. To level the playing field, the government should establish a public-private AI supercomputing center, broadening access for companies and researchers that do not have the adequate funds to conduct the R&D required to advance in AI.

This initiative will allow stakeholders to pool together their resources to utilize and operate the center. The center will allow organization to:

- Schedule compute to run major experiments and model training they would otherwise be incapable of doing.
- Share their research amongst partners.
- Further improve U.S. competitive advantage globally.

An example of this is the NSF NAIRR Pilot that is being performed from January 24, 2024 to January 24, 2026.

We provide two recommendations that will improve innovation, collaboration, and competition:

Community Engagement Programs: To improve transparency and public trust in AI, we also propose community engagement programs integrate with the supercomputing center's operations. These programs would inform the public of the development process by providing educational workshops designed to enhance AI literacy. By doing this the government can help democratize AI technologies and open new opportunities for collaborative innovation.

Contracting Agreements: The government should also consider altering their procurement strategies to align with these goals. The government should prioritize more streamlined contracting agreements for these small businesses, such as Commercial Solutions Openings (CSOs) and Other Transaction Authorities (OTAs). By doing so, the government would allow smaller, innovative companies to compete and collaborate on specific AI components, driving innovation and diversifying the market.